

cells - Answer Key

1. Describe the process of cellular respiration and its stages.

Answer: Cellular respiration is the process by which cells convert nutrients into energy in the form of ATP. It consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain. Glycolysis occurs in the cytoplasm and produces a small amount of ATP. The Krebs cycle occurs in the mitochondrial matrix and produces more ATP. The electron transport chain occurs in the inner mitochondrial membrane and produces most of the cell's ATP through chemiosmosis.

Explanation: Cellular respiration is a critical process by which cells generate energy from nutrients.

2. Explain the structure and function of the cell membrane.

Answer: The cell membrane is a selectively permeable barrier that separates the cell's interior from its external environment. It is composed primarily of a phospholipid bilayer embedded with proteins, as described by the fluid mosaic model. The membrane controls what enters and exits the cell through passive transport and active transport, which requires energy in the form of ATP. The cell membrane plays a critical role in maintaining the cell's internal environment and regulating the movement of substances in and out of the cell.

Explanation: The cell membrane is a critical component of the cell that regulates the movement of substances and maintains the cell's internal environment.